LUCAS DA DALT

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Engineering student in fluid mechanics with a strong applied-mathematics background, aiming at aerospace and astrophysics research. Seeking a 6-week internship, summer 2027.

EDUCATION

INP-ENSEEIHT (Toulouse INP) — Toulouse, France
Engineering degree (Diplôme d'ingénieur), Fluid Mechanics, Energy & Environment — 2026 – 2029

- Master's-level "Grande École" of engineering. Core courses: fluid mechanics, thermodynamics, linear elasticity, scientific computing

Lycée Michel Montaigne — Bordeaux, France
CPGE MPI, final year — 2025 – 2026

Lycée Camille Guérin — Poitiers, France
CPGE MPI — 2023 – 2025

- Intensive preparatory programme in mathematics, physics and computer science for the national engineering-school entrance exams

French Baccalaureate, with high honours (mention Bien) — 2023

- Specialisations: Mathematics, Physics-Chemistry; Computer Science (NSI) in 11th grade

RESEARCH PROJECTS (TIPE)

Anomaly detection in time series — 2025 – 2026

- Built a statistical model of normal behaviour from 12,000 windows of real data (NAB benchmark, 10 series of 15,000 points)
- Combined sliding statistics (Chebyshev bound) and a spectral distance (Wiener–Khinchin theorem) in Python; detected 100% of labelled anomalies on the Apple series

Signal processing with the fast Fourier transform — 2024 – 2025

- Proved the orthogonality of the Fourier basis and the O(N log N) cost of the FFT; implemented a naive DFT in C and a recursive FFT in OCaml
- Cut the cost from 10^6 to 10^4 operations for 1,024 points; extended the method to fast convolution and the fractional Fourier transform

WORK EXPERIENCE

Private tutor, mathematics and physics — 2024 – present

- Tutor students from middle school to first-year CPGE (MP and ECG tracks): diagnose gaps, design exercises, track progress

Babysitter — 2019 – 2024

- Regular childcare alongside secondary school and preparatory classes

CIVIC ENGAGEMENT

Volunteer, emergency reception centre — Bordeaux, July 2026

- Welcomed and guided residents evacuated during the Gironde wildfires (Parc des Expositions)

Octobre Rose inter-INP cross-country race — October 2026

- Ran for breast-cancer awareness with the ENSEEIHT Run & Ride group

SKILLS, ACTIVITIES & INTERESTS

Languages	French (native), English (B2)
Technical	Python (NumPy, SciPy, Pandas, Matplotlib), C, OCaml, LaTeX
Mathematics	Fourier analysis, probability and statistics, algorithmic complexity
Activities	Running (ENSEEIHT Run & Ride); trumpet; 7 years of piano; music theory diploma, level N3 (École Saint-Bruno, Bordeaux)
Interests	Aerospace and astrophysics, reading, music of every genre